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Fluid end assembly

Abstract

A fluid end having its fluid flow bores sealed without threading a retaining nut into the walls of each bore. The fluid ends may be assembled using a plurality of different kits that each comprise a fluid end body, a component, a retainer element, and a fastening system. The retainer element holds the component within each of the bores formed in the fluid end body and the fastening system secures the retainer element to the body. The fastening system comprises a plurality of externally threaded studs, washers and nuts in some embodiments. In other embodiments, the fastening system comprises a plurality of screws.

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Background/Summary

SUMMARY

(1) The invention is directed to a kit comprising a fluid end body, a component, a retainer element, and a fastening system. The body comprises an external surface, and a bore extending through the body and terminating at an opening formed in the external surface. The component is configured for removable installation within the bore, and the retainer element is engageable with the component when the component is installed within the bore. The fastening system is configured to releasably hold the retainer element against the component when the component is installed within

the bore. The bore has no internal threads formed within that portion that surrounds the component when the component is installed within the bore.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a partially exploded view of a first embodiment of a fluid end of the present invention. FIG. **1** shows a suction and discharge end of the fluid end.

(2) FIG. **2** is a partially exploded view of a plunger end of the fluid end body shown in FIG. **1**.

(3) FIG. **3** is a cross-sectional view of the fluid end shown in FIG. **1**, taken along line A-A.

(4) FIG. **4** is a partially exploded view of a second embodiment of a fluid end. FIG. **4** shows a suction and discharge end of the fluid end.

(5) FIG. **5** is a partially exploded view of a plunger end of the fluid end body shown in FIG. **4**.

(6) FIG. **6** is a cross-sectional view of the fluid end shown in FIG. **4**, taken along line B-B.

(7) FIG. **7** is a partially exploded view of a third embodiment of a fluid end. FIG. **7** shows a suction and discharge end of the fluid end.

(8) FIG. **8** is a partially exploded view of a plunger end of the fluid end body shown in FIG. **7**.

(9) FIG. **9** is a partially exploded view of a fifth embodiment of a fluid end. FIG. **9** shows a suction and discharge end of the fluid end.

(10) FIG. **10** is a partially exploded view of a plunger end of the fluid end body shown in FIG. **9**.

(11) FIG. **11** is a cross-sectional view of the fluid end shown in FIG. **9**, taken along line C-C.

(12) FIG. **12** is a partially exploded view of a sixth embodiment of a fluid end. FIG. **12** shows a suction and discharge end of the fluid end.

(13) FIG. **13** is a cross-sectional view of the fluid end shown in FIG. **12**, taken along line D-D.

(14) FIG. **14** is a partially exploded view of a seventh embodiment of a fluid end. FIG. **14** shows a suction and discharge end of the fluid end.

(15) FIG. **15** is a side elevational view of one of the plurality of studs for use with the fluid ends.

DETAILED DESCRIPTION

(16) Fluid end assemblies are typically used in oil and gas operations to deliver highly pressurized corrosive and/or abrasive fluids to piping leading to the wellbore. The assemblies are typically attached to power ends run by engines. The power ends reciprocate plungers within the assemblies to pump fluid throughout the fluid end. Fluid may be pumped through the fluid end at pressures that range from 5,000-15,000 pounds per square inch (psi). Fluid used in high pressure hydraulic fracturing operations is typically pumped through the fluid end at a minimum of 8,000 psi; however, fluid will normally be pumped through the fluid end at pressures around 10,000-15,000 psi during such operations.

(17) In fluid end assemblies known in the art, the fluid flow passages or bores formed within the fluid end body are typically sealed by inserting a plug into each bore. A large retaining nut is then installed into each bore above the plug. The retaining nuts typically thread into internal threads formed in the walls of each bore.

(18) In operation, the high level of fluid pressure pumping throughout the fluid end may cause the retaining nuts to back off or unthread from their installed position. When a retaining nut unthreads from its installed position, the plug it was retaining may be displaced by fluid pressure. Displacement of the plug allows fluid to leak around the plug and erode the walls of the bore. The internal threads formed in the bores for engagement with the retaining nuts are also known to crack over time. Erosion of the bore walls or cracking of the internal threads typically requires repair or replacement of the fluid end.

(19) The present invention is directed to a plurality of different fluid ends having bores sealed without threading retaining nuts into the walls of each bore. As a result, the fluid ends of the

present invention do not have internal threads formed in their bores proximate the bore openings. Removal of the internal threads eliminates the problems associated with the internal thread failures and the retaining nuts becoming unthreaded from the bores.

(20) With reference to FIGS. **1** and **3**, a first embodiment of a fluid end **100** is shown. The fluid end **100** comprises a fluid end body **102** having a flat external surface **104** and a plurality of first and second bores **106**, **108** formed adjacent one another therein, as shown in FIG. **1**. Preferably, the number of first bores **106** equals the number of second bores **108**. More preferably, each first bore **106** intersects its paired second bore **108** within the fluid end body **102** to form an internal chamber **112**, as shown in FIG. **3**.

(21) FIG. **1** shows five first and second bores **106**, **108**. In alternative embodiments, the number of sets of paired first and second bores in the fluid end body may be greater than five, or less than five. Thus, FIG. **4** shows a fluid end body that includes three sets of paired first and second bores. Each bore of each set of paired bores **106** and **108** terminates in a corresponding opening **110** formed in the external surface **104**. The bores **106** and **108** and openings **110** exist in one-to-one relationship. A plurality of internally threaded openings **144** are formed in the body **102** and uniformly spaced around each bore opening **110**, as shown in FIG. **1**.

(22) With reference to FIG. **3**, each second bore **108** may have an intake opening **118** formed proximate the bottom end of the fluid end body **102**. Each intake opening **118** is connected in one-to-one relationship to a corresponding coupler or pipe. These couplers or pipes are fed from a single common piping system (not shown).

(23) A pair of valves **120** and **122** are positioned within each second bore **108**. The valves **120**, **122** route fluid flow within the body **102**. The intake valve **120** blocks fluid backflow through the intake opening **118**. The discharge valve **122** regulates fluid through one or more discharge openings **126**. A plurality of couplers **127** may be attached to each discharge opening **126** for connection to a piping system (not shown), as shown in FIG. **1**.

(24) Continuing with FIGS. **1** and **3**, the fluid end **100** further comprises a plurality of sets of components **128** and **130**. The number of sets preferably equals the number of sets of paired first and second bores **106** and **108** formed in the body **102**. The component **128** is positioned within a first bore **106**, and the component **130** is positioned within its paired second bore **108**. In one embodiment, the component **128** is a suction plug and the component **130** is a discharge plug. Each of the components **128** and **130** are substantially identical in shape and construction, and each is sized to fully block fluid flow within the respective bore **106**, **108**. A seal **136** is positioned around the outer surface of each component **128**, **130** to block fluid from leaking from the bores **106**, **108**.

(25) Each of the components **128** and **130** comprises a first section **138** joined to a second section **140**. The first section **138** has a footprint sized to cover the bore opening **110** and the second section **140** is configured for removable receipt within one of the bores **106**, **108**. In one embodiment, the first section **138** is an enlarged plate and the second section **140** is a plug sized to be closely received within one of the bores **106**, **108**. When the component **128** or **130** is installed within one of the bores **106**, **108**, the first section **138** engages with the external surface **104** of the body **102**. This engagement prevents longitudinal movement of the second section **140** within the bore **106** or **108** as shown in FIG. **3**.

(26) With reference to FIG. **1**, the first section **138** may be formed as a circular structure having a plurality of notches **142** cut from its outer periphery. When each of the first sections **138** is engaged with the external surface **104** of the body **102**, each of the notches **142** partially surrounds one of the openings **144** spaced around each bore opening **110**.

(27) Continuing with FIGS. **1** and **3**, once each component **128**, **130** is installed in the fluid end body **102**, each of the components **128**, **130** is secured in place by a retainer element **132** in a one-to-one relationship. Each retainer element **132** has a footprint sized to fully cover the first section **138** of the components **128** and **130**. The retainer elements **132** shown in FIG. **1** are flat and cylindrical. A plurality of openings **146** are formed about the periphery of each retainer element

132. Each opening **146** is alignable with a corresponding one of the openings **144** in a one-to-one relationship.

(28) Each of the retainer elements **132** is secured to the fluid end body **102** using a fastening system **134**. The fastening system comprises a plurality of studs **148**, a plurality of washers **150**, and a plurality of nuts **152**. Each stud **148** is externally threaded adjacent its first end **149**, while each opening **144** has internal threads that mate with those of the stud **148**. Each stud **148** may be threaded into place within a corresponding one of the openings **144**, in a one-to-one relationship.

(29) Once a first stud **148** has been installed in the body **102** at its first end **149**, its opposed second end **151** projects from the body's external surface **104**. When each component **128** is positioned within its bore **106**, each of its notches **142** at least partially surrounds a corresponding one of the studs **148**. Likewise, when each component **130** is positioned within its bore **108**, each of its notches **142** at least partially surrounds a corresponding one of the studs **148**.

(30) Each peripheral opening **146** formed in each of the retainer elements **132** is registerable with a corresponding one of the studs **148**. The plurality of washers **150** and nuts **152** may be installed and torqued on each one of the studs **148**. The plurality of washers **150** and nuts **152** hold the retainer element **132** against the first section **138** of the components **128**, **130** and hold the first section **138** against the external surface **104** of the fluid end body **102**. Because each of the retainer elements **132** is attached to the fluid end body **102** using the fastening system **134**, no external threads are formed on the outer surface of each retainer element **132**. Likewise, no internal threads are formed within the walls of each bore **106**, **108**.

(31) With reference to FIGS. **2** and **3**, a plunger end **154** of the fluid end **100** is shown. The plurality of first bores **106** terminate at openings **156** formed on the external surface **104** of the plunger end **154**. An internal seat **159** is formed in the walls of each of the bores **106** proximate each of the bore openings **156**. A plurality of threaded openings **161** are formed in each of the internal seats **159**, as shown in FIG. **2**.

(32) A component **158** is positioned within each first bore **106** through each of the openings **156**. Each of the components **158** is tubular and sized to be closely received within each bore **106**. In one embodiment, the components **158** are stuffing box sleeves.

(33) With reference to FIG. **3**, each of the components **158** may have a first section **160** that joins a second section **162** via a tapered section **164**. The first section **160** may have a larger diameter than the second section **162**. When each of the components **158** are installed within each of the bores **106**, the tapered section **164** engages a tapered seat **166** formed in the walls of each bore **106**. This engagement prevents longitudinal movement of each component **158** within each bore **106**. A seal **167** is positioned around the outer surface of the second section **162** of each of the components **158** in order to block fluid from leaking from the bores **106**.

(34) Once installed within the body **102**, each component **158** is secured in place by a retainer element **170** in a one-to-one relationship. Each of the retainer elements **170** is sized to be closely received within each bore **106** and engage a top surface **171** of each component **158**, as shown in FIG. **3**. Each of the retainer elements **170** shown in FIG. **2** has a cylindrical body and a threaded central opening **172**. A plurality of openings **174** are formed about the periphery of each of the retainer elements **170**. The openings **174** are uniformly spaced around each central opening **172**.

(35) A plurality of ports **175** may be formed in an outer surface of each retainer element **170** that are orthogonal to the plurality of openings **174**. At least one seal **176** may also be disposed around the outer surface of each of the retainer elements **170**. The seal **176** helps block fluid from leaking from the bores **106**.

(36) Each of the retainer elements **170** is secured to the fluid end body **102** using a fastening system **178**. The fastening system **178** comprises a plurality of threaded screws **180**. The screws **180** are preferably socket-headed cap screws.

(37) The fastening system **178** secures each retainer element **170** to each internal seat **159**. When each retainer element **170** is positioned within each bore **106**, each of the peripheral openings **174**

is alignable with a corresponding one of the openings **161** in a one-to-one relationship. Each of the screws **180** is registerable within one of the openings **161** in the seat **159** and one of the peripheral openings **174** in the retainer element **170**.

(38) The screws **180** may be torqued as desired to tightly attach each of the retainer elements **170** to each internal seat **159** and securely hold each component **158** within each bore **106**. Because each of the retainer elements **170** is attached to the fluid end body **102** using the fastening system **178**, no external threads are formed on the outer surface of each of the retainer elements **170**. Likewise, no internal threads are formed within the walls of each bore **106** on the plunger end **154** of the body **102**.

(39) Continuing with FIGS. **2-3**, a plurality of packing seals **181** may be positioned within each of the components **158** and each of the retainer elements **170** to prevent fluid from leaking from the bores **106**. At least one of the packing seals **181** may have a plurality of ports **179** formed in its outer periphery, as shown in FIG. **2**. The ports **179** provide an exit for fluid trapped within the packing seals **181**. Fluid exiting the ports **179** may exit the retainer element **170** through the ports **175**.

(40) A packing nut **182** may also be threaded into the central opening **172** of each of the retainer elements **170** in a one-to-one relationship. The packing nut **182** has a threaded section **183** joined to a body **184**. The body **184** shown in FIG. **2** is cylindrical. However, the body **184** may also be square or rectangular shaped. A central passage **185** extends through the threaded section **183** and the body **184**. The threaded section **183** of the packing nut **182** is threaded into the central opening **172** of the retainer element **170**.

(41) When installed within each of the retainer elements **170**, each of the packing nuts **182** engages with and compresses the packing seals **181** installed within each component **158** and retainer element **170**, as shown in FIG. **3**. Compression of the packing seals **181** helps prevent fluid from leaking past the seals **181**. A seal **186** may also be positioned within the central passage **185** of each of the packing nuts **182** to further seal fluid from leaking from the bores **106**.

(42) A plurality of holes **187** are formed around the outer surface of each of the packing nut bodies **184**. The holes **187** serve as connection points for a spanner wrench that may be used to tightly thread the packing nut **182** into the central opening **172** of each of the retainer elements **170**.

(43) A plunger **188** may also be installed within each bore **106** in a one-to-one relationship. When a plunger **188** is installed within a bore **106**, the plunger **188** is positioned within the component **158**, the retainer element **170**, and the packing nut **182**, as shown in FIG. **3**. Each of the plungers **188** projects from the plunger end **154** of the fluid end body **102** and is attached to a separate power end. As discussed above, the power end reciprocates each of the plungers **188** within the fluid end body **102** so as to pump fluid throughout the body. Each of the plungers **188** may be attached to the power end via a clamp **190** in a one-to-one relationship.

(44) Several kits are useful for assembling the fluid end **100**. A first kit comprises a plurality of the components **128** or **130**, a plurality of the retainer elements **132**, and the fastening system **134**. A second kit may comprise the plurality of components **158**, a plurality of the retainer elements **170**, and the fastening system **178**. The second kit may further comprise a plurality of the packing seals **181**, a plurality of the packing nuts **182**, and a plurality of the plungers **188**. Each of the kits may be assembled using the fluid end body **102**.

(45) With reference to FIGS. **4** and **6**, a second embodiment of a fluid end **200** is shown. The fluid end **200** comprises a fluid end body **202** having a flat external surface **204** and a plurality of first and second bores **206**, **208** formed adjacent one another therein, as shown in FIG. **4**. Each bore of each set of paired bores **206** and **208** terminates in a corresponding opening **210** formed in the external surface **204**. A plurality of threaded openings **211** are formed in the body **202** and uniformly spaced around each opening **210**. The internal functions of the fluid end **200** are identical to those described with reference to fluid end **100**, shown in FIG. **3**.

(46) The fluid end **200** further comprises a plurality of sets of components **212** and **214**. The

number of sets preferably equals the number of set of paired first and second bores **206** and **208** formed in the body **202**. The component **212** is positioned within a first bore **206**, and the component **214** is positioned within its paired second bore **208**. In one embodiment, the component **212** is a suction plug and the component **214** is a discharge plug.

(47) Each of the components **212** and **214** is substantially identical in shape and construction, and is sized to fully block fluid flow within the respective bore **206**, **208**. A seal **216** is positioned around the outer surface of each component **212**, **214** to block fluid from leaking from the bores **206**, **208**.

(48) As shown in FIG. **4**, a top surface **213** of each component **212**, **214** may sit flush with the external surface **204** of the body **202** when installed within a respective bore **206**, **208**. Each of the components **212** and **214** may engage with internal seats (not shown) formed in the walls of each of the bores **206**, **208**. Such engagement helps prevent longitudinal movement of the components **212**, **214** within the respective bore **206**, **208**.

(49) Once installed within the fluid end body **202**, each component **212** and **214** is secured in place by a retainer element **218** in a one-to-one relationship. Each of the retainer elements **218** has a footprint sized to cover a single bore opening **210**. The retainer elements **218** shown in FIG. **4** are flat and cylindrical. A plurality of openings **220** are formed about the periphery of each retainer element **218**. Each peripheral opening **220** is alignable with a corresponding one of the openings **211** in a one-to-one relationship, as shown in FIG. **4**.

(50) The retainer elements **218** are secured to the external surface **204** of the fluid end body **202** by a fastening system **222**. The fastening system **222** comprises a plurality of externally threaded studs **224**, a plurality of washers **226**, and a plurality of internally threaded nuts **228**. Each stud **224** is externally threaded adjacent its first end **230**, while each opening **211** has internal threads that mate with those of the stud **224**. Each stud **224** may be threaded into place within a corresponding one of the openings **211**, in a one-to-one relationship.

(51) Once a first stud **224** has been installed in the body **202** at its first end **230**, its opposed second end **232** projects from the body's external surface **204**. Each peripheral opening **220** formed in the retainer elements **218** is registerable with a corresponding one of the studs **224**. The plurality of washers **226** and nuts **228** may be installed and torqued on each of the studs **224**. The plurality of washers **226** and nuts **228** hold the retainer elements **218** against the external surface **204** of the fluid end body **202**. Because each of the retainer elements **218** is attached to the fluid end body **202** using the fastening system **222**, no external threads are formed on the outer surface of each retainer element **218**. Likewise, no internal threads are formed within the walls of each bore **206** and **208**.

(52) With reference to FIGS. **5-6**, a plunger end **234** of the fluid end **200** is shown. The plurality of first bores **206** terminate at openings **236** formed on the external surface **204** of the plunger end **234**. The plunger end **234** of the fluid end body **202** is similar to the plunger end **154** of fluid end body **102**, shown in FIGS. **2-3**, except that an internal seat **159** is not formed within each bore **206**. Instead, a plurality of internally threaded openings **238** are formed in the external surface **204** of the fluid end body **202** that are uniformly spaced around each bore opening **236**.

(53) A component **240** is positioned within each first bore **206** through each of the openings **236** in a one-to-one relationship. Each of the components **240** is tubular and sized to be closely received within each bore **206**. In one embodiment, the components **240** are stuffing box sleeves.

(54) With reference to FIG. **6**, each of the components **240** may have a first section **242** that joins a second section **244** via a tapered section **246**. The first section **242** may have a larger diameter than the second section **244**. When each of the components **240** are installed within each of the bores **206**, the tapered section **246** engages a tapered seat **248** formed in the walls of each bore **206**. This engagement prevents longitudinal movement of each component **240** within each bore **206**. A seal **250** is positioned around the outer surface of the second section **244** of each of the components **240** to block fluid from leaking from the bores **206**.

(55) Once installed within the body **202**, a top surface **252** of each of the components **240** may sit

flush with the external surface **204** of the body **202**. Each of the components **240** is secured in place within each bore **206** by a retainer element **254** in a one-to-one relationship. The retainer elements **254** shown in FIG. **5** have a cylindrical body and a threaded central opening **256**. A plurality of openings **258** are formed about the periphery of each of the retainer elements **254**. The openings **258** are uniformly spaced around each central opening **256**.

(56) The retainer elements **254** are secured to the external surface **204** of the fluid end body **202** using a fastening system **260**. The fastening system **260** comprises a plurality of threaded screws **262**. The screws **262** are preferably socket-headed cap screws. When each retainer element **254** is positioned over each bore opening **236**, each of the peripheral openings **258** is alignable with a corresponding one of the openings **238** in a one-to-one relationship. Each of the screws **262** is registerable within one of the openings **238** in the body **202** and one of the peripheral openings **258** in each of the retainer elements **254**.

(57) The screws **262** may be torqued as desired to tightly attach each of the retainer elements **254** to the body **202** and securely hold each of the components **240** within each bore **206**. Because each of the retainer elements **254** is attached to the fluid end body **202** using the fastening system **260**, no external threads are formed on the outer surface of each retainer element **254**. Likewise, no internal threads are formed within the walls of each bore **206** on the plunger end **234** of the body **202**.

(58) Similar to the plunger end **154** shown in FIG. **2**, a plurality of packing seals **264** may be positioned within each of the components **240**. A packing nut **266** may thread into the central opening **256** of each retainer element **254** and compress the packing seals **264**. A seal **267** may also be positioned within each packing nut **266**. Additionally, a plurality of plungers **268** may be disposed within each component **240**, retainer element **254**, and packing nut **266**. Each of the plungers **268** may be attached to a power end via a clamp **270**.

(59) In alternative embodiments, the components **212**, **214**, and **240** may not be flush with the external surface **204** of the body **202** when installed in the respective bores **206**, **208**. In such case, a flange or ledge may be formed on each of the retainer elements **218** or **254** on its side facing the component **212**, **214**, or **240**. The flange or ledge may be installed within the bores **206**, **208** so that it tightly engages the top surface **213** or **252** of the components **212**, **214**, or **240**.

(60) Likewise, if the components **212**, **214**, or **240** project from the external surface **204** of the body **202** when installed within the respective bores **206**, **208**, the retainer elements **218** or **254** can be modified to accommodate the component **212**, **214**, or **240**. For example, a cut-out may be formed in the retainer element **218** or **254** for closely receiving the portion of the component **212**, **214**, or **240** projecting from the body **202**. The area of the retainer element **218** or **254** surrounding the cut-out will engage the external surface **204** of the body **202**.

(61) Several kits are useful for assembling the fluid end **200**. A first kit comprises a plurality of the components **212** or **214**, a plurality of retainer elements **218**, and the fastening system **222**. A second kit may comprise the plurality of components **240**, a plurality of the retainer elements **254**, and the fastening system **260**. The second kit may further comprise a plurality of packing seals **264**, a plurality of packing nuts **266**, and a plurality of plungers **268**. Each of the kits may be assembled using the fluid end body **202**.

(62) Turning now to FIG. **7**, a third embodiment of a fluid end **300** is shown. The fluid end **300** comprises a fluid end body **302** having a flat external surface **304** and a plurality of first and second bores **306**, **308** formed adjacent one another therein. Each bore of each set of paired bores **306** and **308** terminates in a corresponding opening **310** formed in the external surface **304**. A plurality of threaded openings **311** are formed in the body **302** and uniformly spaced around each bore opening **310**. The internal functions of the fluid end **300** are identical to those described with reference to fluid end **100**, shown in FIG. **3**.

(63) The fluid end **300** further comprises a plurality of sets of components **312** and **314**. The number of sets preferably equals the number of set of paired first and second bores **306** and **308** formed in the body **302**. The component **312** is positioned within a first bore **306**, and the

component **314** is positioned within its paired second bore **308**. In one embodiment, the component **312** is a suction plug and the component **314** is a discharge plug. A seal **315** is positioned around each of the components **312**, **314** to block fluid from leaking from the respective bores **306**, **308**.
(64) The components **312** and **314** have the same shape and construction as the components **212** and **214** shown in FIGS. **4** and **6**. Each of the components **312** and **314** may engage with internal seats (not shown) formed in the walls of each of the bores **306**, **308**. Such engagement helps prevent longitudinal movement of the components **312**, **314** within the respective bores **306**, **308**.
(65) Once installed within the body **302**, a top surface **313** of each of the components **312**, **314** may sit flush with the external surface **304** of the body **302**. Each of the components **312**, **314** is secured within each respective bore **306**, **308** by a retainer element **316**. Each of the retainer elements **316** shown in FIG. **7** is a large rectangular plate having a footprint sized to cover a plurality of adjacent bore openings **310** at one time. A plurality of openings **318** are formed in each retainer element **316** that are alignable with a corresponding one of the openings **311** in a one-to-one relationship.
(66) Each of the retainer elements **316** is secured to the external surface **304** of the fluid end body **302** by a fastening system **320**. The fastening system **320** comprises a plurality of externally threaded studs **322**, a plurality of washers **324**, and a plurality of internally threaded nuts **326**. The fastening system **320** secures each of the retainer elements **316** on the fluid end body **302** in the same way as described with reference to the fastening system **222** used with the fluid end **200**.
(67) Because each of the retainer elements **316** is attached to the fluid end body **302** using the fastening system **320**, no external threads are formed in the retainer element **316**. Likewise, no internal threads are formed within the walls of each bore **306** and **308**.
(68) When the retainer elements **316** are installed on the fluid end body **302**, the edges of the retainer element **316** may extend far enough so as to sit flush with the edges of the fluid end body **302**. In alternative embodiments, the retainer element **316** may have different shapes or sizes. For example, the retainer element **316** may be large enough so as to cover an entire side surface of the fluid end body **302**. Alternatively, the retainer elements **316** may have rounded edges, as shown in FIG. **8**.
(69) Turning to FIG. **8**, a plunger end **330** of the fluid end **300** is shown. The plurality of first bores **306** terminate at openings **332** formed on the external surface **304** of the plunger end **330**. A plurality of internally threaded openings **334** are formed in the external surface **304** that are uniformly spaced around each bore opening **332**.
(70) A component **336** is positioned within each first bore **306** through each of the openings **332**. Each of the components **336** is tubular and sized to be closely received within each bore **306**. In one embodiment, the components **336** are stuffing box sleeves. The components **336** have the same shape and construction as the components **240**, shown in FIGS. **5-6**.
(71) Once installed within the body **302**, a top surface **346** of each of the components **336** may sit flush with the external surface **304** of the body **302**. Each of the components **336** is secured within each bore **306** by a single retainer element **348**. The retainer element **348** shown in FIG. **8** is a large oval plate having a footprint sized to cover a plurality of adjacent bore openings **332** formed on the plunger end **330** of the fluid end body **302**. A plurality of openings **350** are formed in the retainer element **348** that are alignable with a corresponding one of the openings **334** in a one-to-one relationship.
(72) In alternative embodiments, the retainer element **348** may have different shapes or sizes. For example, the retainer element **348** may be large enough so as to cover an entire side surface of the fluid end body **302**. Alternatively, the retainer element **348** may have squared edges, as shown in FIG. **7**.
(73) The retainer element **348** is secured to the external surface **304** of the fluid end body **302** by a fastening system **352**. The fastening system **352** comprises a plurality of screws **354**. The fastening system **352** secures the retainer element **348** on the fluid end body **302** in the same way as described with reference to the fastening system **260** used with the fluid end **200** and shown in

FIGS. **5-6**.

(74) Because the retainer element **348** is attached to the fluid end body **302** using the fastening system **352**, no external threads are formed in the retainer element **348**. Likewise, no internal threads are formed within the walls of each bore **306**.

(75) A central threaded opening **356** is formed in the center of each grouping of openings **350** in the retainer element **348**. The openings **356** are alignable with each bore opening **332** in a one-to-one relationship. A single packing nut **358** may thread into each central opening **356**. A seal **359** may be positioned within each packing nut **358**.

(76) Similar to the plunger end **234** shown in FIGS. **5-6**, a plurality of packing seals **360** may be positioned within each component **336**. Each of the packing nuts **358** may compress the packing seals **360** when installed within the retainer element **348**. A plurality of plungers **362** may be disposed within each component **336**, the retainer element **348**, and each packing nut **358**. Each of the plungers **362** may be connected to a power end via a clamp **364**. A cross-sectional view of the fluid end **300** looks identical to the cross-sectional view of the fluid end **200**, shown in FIG. **6**.

(77) Several kits are useful for assembling the fluid end **300**. A first kit comprises a plurality of the components **312** or **314**, a retainer element **316**, and the fastening system **320**. A second kit may comprise a plurality of the components **336**, a retainer element **348**, and the fastening system **352**. The second kit may further comprise a plurality of the packing seals **360**, a plurality of the packing nuts **358**, and a plurality of the plungers **362**. Each of the kits may be assembled using the fluid end body **302**.

(78) With reference to FIGS. **9** and **11**, a fourth embodiment of a fluid end **400** is shown. The fluid end **400** comprises a fluid end body **402** having a flat external surface **404** and a plurality of first and second bores **406**, **408** formed adjacent one another therein, as shown in FIG. **9**. Each bore of each set of paired bores **406** and **408** terminates in a corresponding opening **410** formed in the external surface **404**. A plurality of threaded openings **411** are formed in the body **402** and uniformly spaced around each opening **410**. The internal functions of the fluid end **400** are identical to those described with reference to fluid end **100**, shown in FIG. **3**.

(79) The fluid end **400** further comprises a plurality of sets of components **412** and **414**. The number of sets preferably equals the number of set of paired first and second bores **406** and **408** formed in the body **402**. The component **412** is positioned within a first bore **406**, and the component **414** is positioned within its paired second bore **408**. In one embodiment, the component **412** is a suction plug and the component **414** is a discharge plug. A seal **415** is positioned around the outer surface of each of the components **412**, **414** to block fluid from leaking from the respective bores **406**, **408**.

(80) The components **412** and **414** have substantially the same shape and construction as the components **212** and **214** shown in FIGS. **4** and **6**. However, in contrast to the components **212**, **214**, each of the components **412** and **414** is joined to a single retainer element **416**.

(81) The components **412**, **414** may be welded or fastened to the center of the back surface of each retainer element **416**. Alternatively, each of the components **412** or **414** and a corresponding retainer element **416** may be machined as a single piece, as shown in FIG. **11**. Each of the retainer elements **416** secures each of the components **412**, **414** within the respective bores **406**, **408**. The retainer elements **416** also prevent the components **412**, **414** from moving longitudinally within the respective bores **406**, **408**.

(82) A plurality of openings **418** are formed about the periphery of each retainer element **416**. Each peripheral opening **418** is alignable with a corresponding one of the openings **411** in a one-to-one relationship, as shown in FIG. **9**.

(83) The retainer elements **416** are secured to the external surface **404** of the body **402** using a fastening system **420**. The fastening system **420** comprises a plurality of externally threaded studs **422**, a plurality of washers **424**, and a plurality of internally threaded nuts **426**. The fastening system **420** secures the retainer elements **416** to the fluid end body **402** in the same way as

described with reference to the fastening system **222** used with the fluid end **200**.
(84) Because the retainer elements **416** are attached to the fluid end body **402** using the fastening system **420**, no external threads are formed in the retainer elements **416**. Likewise, no internal threads are formed within the walls of each bore **406** and **408**.
(85) Turning now to FIGS. **10-11**, a plunger end **430** of the fluid end **400** is shown. The plurality of first bores **406** terminate at openings **432** formed on the external surface **404** of the plunger end **430**. A plurality of internally threaded openings **434** are formed in the external surface **404** that are uniformly spaced around each bore opening **432**.
(86) A component **436** is positioned within each first bore **406** through each of the openings **432**. Each of the components **436** is tubular and sized to be closely received within each bore **406**. In one embodiment, the components **436** are stuffing box sleeves. The components **436** have substantially the same shape and construction as the components **240**, shown in FIGS. **5-6**. However, in contrast to the components **240**, each of the components **436** is joined to a single retainer element **438**.
(87) The components **436** may be welded or fastened to the center of the back surface of each retainer element **438**. Alternatively, each of the components **436** and a corresponding retainer element **438** may be machined as a single piece, as shown in FIG. **11**. Each of the retainer elements **438** secures each of the components **436** within the bores **406**. The retainer elements **438** also prevent the components **436** from moving longitudinally within the bores **406**.
(88) A threaded central opening **440** is formed within each retainer element **438**. A plurality of threaded openings **442** are formed about the periphery of each of the retainer elements **438** and are uniformly spaced around each central opening **440**. Each peripheral opening **442** is alignable with a corresponding one of the openings **434** in a one-to-one relationship, as shown in FIG. **10**.
(89) The retainer elements **438** are secured to the external surface **404** of the body **402** using a fastening system **444**. The fastening system **444** comprises a plurality of screws **446**. The fastening system **444** secures the retainer elements **438** to the fluid end body **402** in the same way as described with reference to the fastening system **260** used with the fluid end **200** and shown in FIGS. **5-6**.
(90) Because the retainer elements **438** are attached to the fluid end body **402** using the fastening system **444**, no external threads are formed in the retainer elements **416**. Likewise, no internal threads are formed within the walls of each bore **406** on the plunger end **430** of the body **402**.
(91) Like the plunger end **330** of fluid end **300**, the fluid end **400** may also comprise a plurality of packing seals **448**, a plurality of packing nuts **450**, each housing a seal **454**, and a plurality of plungers **456**. Each plunger **456** may be connected to a power end via a clamp **458**.
(92) Several kits are useful for assembling the fluid end **400**. A first kit comprises a plurality of the components **412** or **414**, a plurality of the retainer elements **416**, and the fastening system **420**. A second kit may comprise a plurality of the components **436**, a plurality of the retainer elements **438**, and the fastening system **444**. The second kit may further comprise a plurality of the packing seals **448**, a plurality of the packing nuts **450** and a plurality of the plungers **456**. Each of the kits may be assembled using the fluid end body **402**.
(93) With reference to FIGS. **12-13**, a fifth embodiment of a fluid end **500** is shown. The fluid end **500** comprises a fluid end body **502** having a flat external surface **504** and a plurality of first and second bores **506**, **508** formed adjacent one another therein, as shown in FIG. **12**. Each bore of each set of paired bores **506** and **508** terminates in a corresponding opening **510** formed in the external surface **504**. A plurality of threaded openings **511** are formed in the body **502** and uniformly spaced around each opening **510**. The internal functions of the fluid end **500** are identical to those described with reference to fluid end **100**, shown in FIG. **3**.
(94) The fluid end **500** further comprises a plurality of sets of components **512** and **514**. The number of sets preferably equals the number of set of paired first and second bores **506** and **508** formed in the body **502**. The component **512** is positioned within a first bore **506**, and the

component **514** is positioned within its paired second bore **508**. In one embodiment, the component **512** is a suction plug and the component **514** is a discharge plug. The components **512** and **514** have the same shape and construction as the components **212** and **214** shown in FIGS. **4** and **6**. A seal **516** is positioned around the outer surface of each component **512**, **514** to block fluid from leaking from the bores **506**, **508**.

(95) As shown in FIG. **12**, a top surface **513** of each of the components **512**, **514** may sit flush with the external surface **504** of the body **502** when installed within a respective bore **506**, **508**. Each of the components **512** and **514** may engage with internal seats (not shown) formed in the walls of each of the bores **506**, **508**. Such engagement helps prevent longitudinal movement of the components **512**, **514** within the respective bore **506**, **508**.

(96) Once installed within the fluid end body **502**, each component **512** and **514** is secured in place by a retainer element **518** in a one-to-one relationship. Each of the retainer elements **518** has a footprint sized to cover a single bore opening **510**. The retainer elements **518** shown in FIG. **12** are flat and cylindrical and each have a central threaded opening **519**. A plurality of openings **520** are formed about the periphery of each retainer element **518** and are uniformly spaced around each central opening **519**. Each peripheral opening **520** is alignable with a corresponding one of the openings **511** in a one-to-one relationship, as shown in FIG. **12**.

(97) The retainer elements **518** are secured to the external surface **504** of the fluid end body **502** by a fastening system **522**. The fastening system **522** comprises a plurality of externally threaded studs **524**, a plurality of washers **526**, and a plurality of internally threaded nuts **528**. The fastening system **522** secures the retainer elements **518** to the fluid end body **502** in the same way as described with reference to the fastening system **222** used with the fluid end **200** shown in FIGS. **4** and **6**.

(98) Each central opening **519** formed in each retainer element **518** is alignable with each corresponding bore opening **510** in a one-to-one relationship. A retaining nut **530** may thread into each central opening **519** to cover each bore opening **510**. Using a threaded retaining nut **530** with the retainer element **518** allows access to each bore opening **510** without having to remove the retainer elements **518** from the fluid end body **502**.

(99) While the fluid end **500** uses a threaded retaining nut **530**, the retaining nut **530** is not threaded into the walls of the bores **506**, **508**. Thus, any failures associated with the retaining nut **530** may be experienced in the retainer element **518**, which is easily replaceable. This similar configuration is used on the plunger end **234** of the fluid end **200** shown in FIGS. **5-6**. Such configuration is shown again on a plunger end **532** of the fluid end body **502** in FIG. **13**.

(100) A kit is useful for assembling the fluid end **500**. The kit may comprise a plurality of the components **512** or **514**, a plurality of the retainer elements **518**, and the fastening system **522**. The kit may further comprise a plurality of retaining nuts **530**. The kit may be assembled using the fluid end body **502**.

(101) Turning now to FIG. **14**, a sixth embodiment of a fluid end **600** is shown. The fluid end **600** comprises a fluid end body **602** having a flat external surface **604** and a plurality of first bores (not shown) and second bores **608** formed adjacent one another therein. Each bore of each set of paired bores terminates in a corresponding opening **610** formed in the external surface **604**. A plurality of threaded openings **611** are formed in the body **602** and uniformly spaced around each opening **610**. The internal functions of the fluid end **600** are identical to those described with reference to fluid end **100**, shown in FIG. **3**.

(102) The fluid end **600** further comprises a plurality of sets of components **614**. The component **614** is positioned within a second bore **608**. The components positioned within each first bore are not shown in FIG. **14**. However, such components are identical in shape and construction to the components **614**.

(103) The number of sets of components preferably equals the number of sets of paired first bores (not shown) and second bores **608** formed in the body **602**. In one embodiment, the component **614**

positioned within a first bore is a suction plug, and the component **614** positioned within its paired second bore **608** is a discharge plug. The components **614** have a substantially similar shape and construction as the components **212** and **214** shown in FIGS. **4** and **6**, except that a threaded hole **616** is formed in a top surface **613** of each component **614**. A seal **618** is positioned around the outer surface of each component **614** to block fluid from leaking from the bores **608**.

(104) The top surface **613** of each component **614** may sit flush with the external surface **604** of the body **602** when installed within a bore **608**. Each of the components **614** may engage with internal seats (not shown) formed in the walls of each of the bores **608**. This engagement helps prevent longitudinal movement of the components **614** within the bore **608**. Likewise, the components positioned within the first bores (not shown) may engage internal seats formed within the walls of the first bores.

(105) Once installed within the fluid end body **602**, each component **614** is secured by a retainer element **620** in a one-to-one relationship. Likewise, the components positioned within the first bores (not shown) are each secured by one of the retainer elements **620**. Each of the retainer elements **620** has a footprint sized to cover a single bore opening **610**. The retainer elements **620** shown in FIG. **14** are flat and cylindrical and each have a central threaded opening **622**. A plurality of openings **624** are formed about the periphery of each retainer element **620** and are uniformly spaced around each central opening **622**. Each peripheral opening **624** is alignable with a corresponding one of the openings **611** in a one-to-one relationship.

(106) The retainer elements **620** are secured to the external surface **604** of the fluid end body **602** by a fastening system **626**. The fastening system **626** comprises a plurality of externally threaded studs **628**, a plurality of washers (not shown), and a plurality of internally threaded nuts **630**. The fastening system **626** secures the retainer elements **620** to the fluid end body **602** in the same way as described with reference to the fastening system **222** used with the fluid end **200** shown in FIGS. **4** and **6**.

(107) The fastening system **626** may further comprise a plurality of eye bolts **632**, a plurality of handles **634**, and a cable **636**. Each eye bolt **632** has external threads **638** formed on its first end and an eye **640** formed on its opposite second end. The threaded end **638** of each eye bolt **632** threads into each hole **616** formed in each component **614** in a one-to-one relationship. Once installed within each hole **616**, the eye **640** of each eyebolt **632** projects through the central opening **622** formed in each retainer element **620**.

(108) Each of the handles **634** has a threaded section **642** joined to a cylindrical body **644**. A central passage **646** extends through the threaded section **642** and the body **644**. Each of the threaded sections **642** may be installed within the central opening **622** of each of the retainer elements **620** such that each eye bolt **632** is disposed within the central passage **646**. Once one of the handles **634** is installed in a retainer element **620**, the eye bolt **632** projects from the handle **634**. The handle **634** helps support the eye bolt **632** and provides a grip to assist in installation or removal of a retainer element **620** on the fluid end body **602**.

(109) The cable **636** may be disposed through each eye **640** of each eye bolt **632**. Each of the eye bolts **632** may be oriented to facilitate the passage of the cable **636** through each eye **640**. The ends of the cable **636** may be attached to the external surface **604** of the fluid end body **602** using eye bolts **650** and clamps **652**. The cable **636** is preferably made of a stiff and tough material, such as high-strength nylon or steel.

(110) In operation, the eyebolts **632** and cable **636** tether each of the retaining elements **620** and components **614**, in case of failure of the retainer elements **620**, a portion of the fastening system **626**, or the fluid end body **602**. When a failure occurs, the large pressure in the fluid end body **602** will tend to force the components **614** out of their respective bores **608** with a large amount of energy. The cable **636** helps to retain the components **614** within the bores **608** in the event of a failure. The cable **636** also helps to retain the retainer elements **620** in position in the event of a failure. The fastening systems **134**, **222**, **320**, **420**, and **522** used with fluid ends **100**, **200**, **300**, **400**,

and **500** may also be configured for use with the eye bolts **632**, handles **634** and cable **636**.
(111) In alternative embodiments, the handles **634** may not be used. A single eye bolt **632** may also be formed integral with a single component **614**. A single cable **636** may also be used through each of the eyebolts **632**. Each cable **636** would independently attach to the external surface **604** of the fluid end body **602**.
(112) Several kits are useful for assembling the fluid end **600**. A first kit comprises a plurality of the components **614**, a plurality of the retainer elements **620**, and the fastening system **626**. The kit may be assembled using the fluid end body **602**.
(113) With reference to FIGS. **1-14**, a single fluid end body may use any combination of the kits described herein. The fluid end bodies, components, and retainer elements described herein are preferably made of high strength steel.
(114) While the fluid end bodies **102**, **202**, **302**, **402**, and **502** shown in FIGS. **1-13** are substantially rectangular in shape, the kits described herein may also be used with any shape of a fluid end body, such as that shown in FIG. **14**. Likewise, the retainer elements described herein may vary in shape and size, as desired. For example, the circular retainer elements described herein may be square or rectangular shaped.
(115) The fastening systems **134**, **222**, **320**, **420**, and **522** described herein each use eight studs around each bore opening. In alternative embodiments, more than eight studs or less than eight studs may be used to secure each retainer element over each bore opening. For example, FIG. **14** only shows six studs securing each retainer element **620** over each bore opening **610**. Likewise, fewer than 16 or more than 16 screws may be used with the fastening systems **178**, **260**, **352**, and **444**. The number of peripheral openings formed in each retainer element described herein may correspond with the number of openings formed around each bore opening in each fluid end body and the number of studs or screws being used.
(116) The fastening systems described herein reduce the amount of torque required to secure each retainer element to the fluid end bodies. Rather than having to torque one large retaining nut, the torque is distributed throughout the plurality of studs, nuts, or screws. Decreasing the amount of torque required to seal the bores increases the safety of the assembly process.
(117) Turning to FIG. **15**, a stud **700** is shown. The stud **700** may be used with the fastening systems **134**, **222**, **320**, **420**, **522**, and **626** shown in FIGS. **1**, **4**, **7**, **9**, **11**, and **14**. For exemplary purposes, the stud **700** will be described with reference to fluid end **100**, shown in FIG. **1**.
(118) The stud **700** has a first threaded section **702** and an opposite second threaded section **704**. The threaded sections **702** and **704** are joined by an elongate, cylindrical body **706**. The first threaded section **702** is configured for threading into one of the plurality of threaded openings **144** formed in the fluid end body **102**. The second threaded section **704** is configured for threading into the threaded opening formed in one of the nuts **152**.
(119) The first section **702** may have fewer threads than that of the opening **144**. For example, if the opening **144** has 18 internal threads, the first section **702** of the stud **700** may only have 16 external threads. This configuration ensures that all of the threads formed on the first section **702** will be engaged and loaded when the first end **702** is threaded into the opening **144**. Engaging all of the threads helps increase the fatigue life of the first end **702** of the stud **700**. Likewise, the second section **704** may have fewer external threads than there are internal threads formed in the nut **152**. The stud **700** may also be subjected to shot peening on its non-threaded sections prior to its use to help reduce the possibility of fatigue cracks. The stud **700** may have a smooth outer surface prior to performing shot peening operations.
(120) The body **706** of the stud **700** comprises a first section **708** and a second section **710**. The first section **708** has a smaller diameter than the second section **710**. The retainer element **132** is primarily held on the first section **708** of the stud **700**. The diameter of the second section **710** is enlarged so that it may center the washer **150** on the stud **700**.
(121) The diameter of the second section **710** is configured so that it is only slightly smaller than

the diameter of the central opening of the washer **150**. This sizing allows the washer **150** to closely receive the second section **710** of the stud **700** when the washer **150** is positioned on the stud **700**. When the washer **150** is positioned on the second section **710**, the washer **150** is effectively centered on the stud **700**. The washer **150** is also effectively centered against the nut **152** once the nut **152** is installed on the stud **700**.

(122) Without placing the washer **150** on the second section **710**, the washer may have to be manually centered on the stud **700** prior to installing the nut **152**. If the washer **150** is not properly centered on the stud **700** or against the nut **152**, it may be difficult to effectively torque or un-torque the nut **152** from the stud **700**, depending on the type of washer used.

(123) The plurality of washers used with each fastening system **134**, **222**, **320**, **420**, **522**, and **626** shown in FIGS. **1**, **4**, **7**, **9**, **11**, and **14** may be configured to allow a large amount of torque to be imposed on the nuts used with the washers without using a reaction arm. Instead, the washer itself may serve as the counterforce needed to torque a nut onto a stud. Not having to use a reaction arm increases the safety of the assembly process. The nuts used with the fastening systems **134**, **222**, **320**, **420**, **522**, and **626** may also comprise a hardened inner layer to help reduce galling between the threads of the nuts and studs during the assembly process. An example of the above described washers, nuts, and methods are described in Patent Cooperation Treaty Application Serial No. PCT/US2017/020548, authored by Junkers, et al., the entirety of which is incorporated herein by reference.

(124) The various features and alternative details of construction of the apparatuses described herein for the practice of the present technology will readily occur to the skilled artisan in view of the foregoing discussion, and it is to be understood that even though numerous characteristics and advantages of various embodiments of the present technology have been set forth in the foregoing description, together with details of the structure and function of various embodiments of the technology, this detailed description is illustrative only, and changes may be made in detail, especially in matters of structure and arrangements of parts within the principles of the present technology to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed.

Claims

1. A hydraulic fluid pump, comprising: a fluid end, comprising: a housing including a first end and a second end opposite the first end, the housing having a plurality of housing bores extending from the first end to the second end and a plurality of valve bores situated orthogonal to the plurality of housing bores; a first plate fastened to the housing at its first end, the first plate having at least one plate bore axially aligned with a selected one of the plurality of housing bores; a plug at least partially disposed within a selected one of the plurality of housing bores and closing the selected one of the plurality of housing bores; a fastener, threaded to the first plate at the at least one plate bore and retaining the plug within the housing bore; a packing configured to surround at least a portion of a reciprocating plunger; a second plate fastened to the housing at its second end, the second plate having at least one plate bore axially aligned with the selected one of the plurality of housing bores; and a component disposed about the packing and retained by the second plate such that the component engages the housing; wherein the second plate has the same number of plate bores as the number of housing bores of the housing.
2. The hydraulic fluid pump of claim 1, in which the first plate has the same number of plate bores as the number of housing bores of the housing.
3. The hydraulic fluid pump of claim 1, in which the fluid end further comprises a plurality of plungers; in which one of the plurality of plungers is the reciprocating plunger of claim 1; and in which each of the plurality of plungers is at least partially disposed within a corresponding housing bore.

4. The hydraulic fluid pump of claim 1, further comprising a packing nut configured to engage the packing such that the packing nut retains the packing within the component.
5. The hydraulic fluid pump of claim 4, in which the packing nut comprises a central opening configured to receive the reciprocating plunger.
6. The hydraulic fluid pump of claim 1 further comprising a valve pair situated in a selected one of the plurality of valve bores.
7. The hydraulic fluid pump of claim 1, further comprising: a plug situated in the selected one of the plurality of valve bores; and an annular seal situated around the plug, the annular seal configured to block fluid from passing by an external surface of the plug.
8. A hydraulic fluid pump, comprising: a fluid end, comprising: a housing including a first end and a second end opposite the first end, the housing having a plurality of housing bores extending from the first end to the second end; a first plate fastened to the housing at its first end, the first plate having at least one plate bore axially aligned with a selected one of the plurality of housing bores; a plug at least partially disposed within a selected one of the plurality of housing bores and closing the selected one of the plurality of housing bores; a fastener, threaded to the first plate at the at least one plate bore and retaining the plug within the housing bore; a packing configured to surround at least a portion of a reciprocating plunger; a second plate fastened to the housing at its second end, the second plate having at least one plate bore axially aligned with the selected one of the plurality of housing bores; a component disposed about the packing and retained by the second plate such that the component engages the housing; wherein the second plate has the same number of plate bores as the number of housing bores of the housing; and a power end, the power end configured to drive the reciprocating plunger.
9. A hydraulic fluid pump, comprising: a fluid end, comprising: a housing including a first end and a second end opposite the first end, the housing having a plurality of housing bores extending from the first end to the second end; a first plate fastened to the housing at its first end, the first plate having at least one plate bore axially aligned with a selected one of the plurality of housing bores; a plug at least partially disposed within the selected one of the plurality of housing bores and closing the selected one of the plurality of housing bores; a packing configured to engage a reciprocating plunger; a second plate fastened to the housing at its second end, the second plate having at least one plate bore axially aligned with the selected one of the plurality of housing bores; and a component disposed about the packing and retained to the housing by the second plate; in which the second plate has the same number of plate bores as the number of housing bores of the housing; and a power end connected to the fluid end.
10. The hydraulic fluid pump of claim 9, in which the housing is a single integrally-formed piece.
11. The hydraulic fluid pump of claim 9, in which the first plate has the same number of plate bores as the number of housing bores of the housing.
12. The hydraulic fluid pump of claim 9, in which the second plate has rounded edges.
13. The hydraulic fluid pump of claim 9, in which the housing comprises a plurality of valve bores situated perpendicular to the plurality of housing bores.